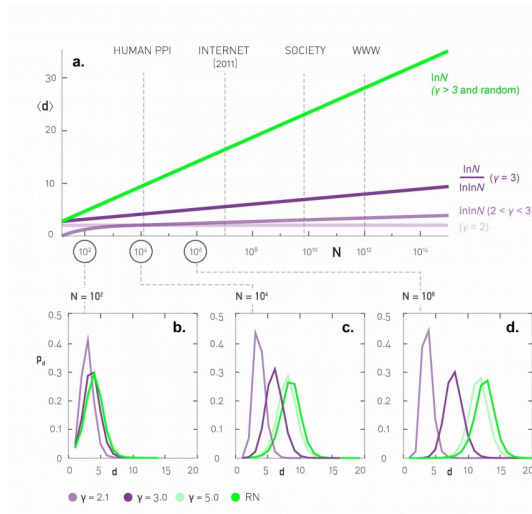


L5: Path Lengths in Power-law Networks



What about power-law networks with other exponent values? How does the CPL of those networks scale with network size?

As we saw in the previous page, when the exponent (shown as γ in this plot) is equal to 3, we get the $O\left(\frac{\ln N}{\ln \ln N}\right)$ expression of PA networks.

The value $\gamma = 3$ is critical because the variance of the degree distribution becomes well-defined when $\gamma > 3$. In that case the power-law networks do not differ from $G(N, p)$ random graphs in terms of their CPL – the average shortest path length increases as $O(\ln N)$.

For γ values between 2 and 3 (i.e., the mean is finite but the variance of the degree distribution diverges), the CPL scales even slower, following a double-log pattern: $O(\ln \ln N)$. These networks are sometimes referred to as **ultra-small world** networks.

The plots at the lower part of the visualization show shortest-path length distributions for three different exponent values as well as for a corresponding $G(N, p)$ random graph. As you see, the differences between all these networks are minor when the network is only a few hundreds of nodes. For networks with millions of nodes, however, we see a major difference in the resulting path length distributions, showing clearly the major effect of the degree distribution exponent as well as the critical value of $\gamma = 3$.

Food For Thought

Look at the literature for a derivation of the previous formula for the CPL of networks when γ is between 2 and 3.